Question 1:

EXPLORER

ASSIGNMENT

- addtwonumbers.class
- addtwonumbers.java 1
- evenorodd.java
- factorialnum.java
- fibonacciseries.java
- largest3num.java 1
- multiplicationtable.java
- nameprint.java
- reversedigits.java
- sumnaturalnum.java 1
- swapnum.java 1

addtwonumbers.java 1

addtwonumbers.java > addtwonumbers

```java
import java.util.Scanner;

public class addtwonumbers {

    Run | Debug
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);
        System.out.print(s: "Enter the 1st number: ");
        int num1 = input.nextInt();
        System.out.print(s: "Enter the 2nd number: ");
        int num2 = input.nextInt();
        int sum = num1 + num2;
        System.out.println("Sum: " + sum);
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CDAC\assignment> javac ./addtwonumbers.java
PS D:\CDAC\assignment> java addtwonumbers
Enter the 1st number: 12
Enter the 2nd number: 14
Sum: 26
PS D:\CDAC\assignment>
```

EXPLORER

ASSIGNMENT

- addtwonumbers.class
- addtwonumbers.java 1
- evenorodd.java
- factorialnum.java
- fibonacciseries.java
- largest3num.java 1
- multiplicationtable.java
- nameprint.java
- reversedigits.java
- sumnaturalnum.java 1
- swapnum.java 1

OUTLINE

TIMELINE

JAVA PROJECTS

swapnum.java 1 fibonacciseries.java

fibonacciseries.java > fibonacciseries

```java
public class fibonacciseries {
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Run: swapnum

```
PS D:\CDAC\assignment> javap addtwonumbers
Compiled from "addtwonumbers.java"
public class addtwonumbers {
  public addtwonumbers();
  public static void main(java.lang.String[]);
}
PS D:\CDAC\assignment> javap -c addtwonumbers
Compiled from "addtwonumbers.java"
public class addtwonumbers {
  public addtwonumbers();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: new           #2                  // class java/util/Scanner
       3: dup
       4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
       7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
      10: astore_1
      11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      14: ldc           #6                  // String Enter the 1st number:
      16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
      19: aload_1
      20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
      23: istore_2
      24: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      27: ldc           #9                  // String Enter the 2nd number:
      29: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
      32: aload_1
      33: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
      36: istore_3
      37: iload_2
      38: iload_3
      39: iadd
      40: istore        4
      42: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      45: new           #10                 // class java/lang/StringBuilder
      48: dup
      49: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
```

factorialnum.java
fibonacciseries.java
largest3num.java 1
multiplicationtable.java
nameprint.java
reversedigits.java
sumnaturalnum.java 1
swapnum.java 1

OUTLINE
TIMELINE
JAVA PROJECTS

```
PS D:\CDAC\assignment> javap -v addtwonumbers
   #22 = Utf8               LineNumberTable
   #23 = Utf8               main
   #24 = Utf8               ([Ljava/lang/String;)V
   #25 = Utf8               SourceFile
   #26 = Utf8               addtwonumbers.java
   #27 = NameAndType        #19:#20        // "<init>":()V
   #28 = Utf8               java/util/Scanner
   #29 = Class              #46            // java/lang/System
   #30 = NameAndType        #47:#48        // in:Ljava/io/InputStream;
   #31 = NameAndType        #19:#49        // "<init>":(Ljava/io/InputStream;)V
   #32 = NameAndType        #50:#51        // out:Ljava/io/PrintStream;
   #33 = Utf8               Enter the 1st number:
   #34 = Class              #52            // java/io/PrintStream
   #35 = NameAndType        #53:#54        // print:(Ljava/lang/String;)V
   #36 = NameAndType        #55:#56        // nextInt:()I
   #37 = Utf8               Enter the 2nd number:
   #38 = Utf8               java/lang/StringBuilder
   #39 = Utf8               Sum:
   #40 = NameAndType        #57:#58        // append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
   #41 = NameAndType        #57:#59        // append:(I)Ljava/lang/StringBuilder;
   #42 = NameAndType        #60:#61        // toString:()Ljava/lang/String;
   #43 = NameAndType        #62:#54        // println:(Ljava/lang/String;)V
   #44 = Utf8               addtwonumbers
   #45 = Utf8               java/lang/Object
   #46 = Utf8               java/lang/System
   #47 = Utf8               in
   #48 = Utf8               Ljava/io/InputStream;
   #49 = Utf8               (Ljava/io/InputStream;)V
   #50 = Utf8               out
   #51 = Utf8               Ljava/io/PrintStream;
   #52 = Utf8               java/io/PrintStream
   #53 = Utf8               print
   #54 = Utf8               (Ljava/lang/String;)V
   #55 = Utf8               nextInt
   #56 = Utf8               ()I
   #57 = Utf8               append
   #58 = Utf8               (Ljava/lang/String;)Ljava/lang/StringBuilder;
   #59 = Utf8               (I)Ljava/lang/StringBuilder;
   #60 = Utf8               toString
   #61 = Utf8               ()Ljava/lang/String;
   #62 = Utf8               println
```

factorialnum.java
fibonacciseries.java
largest3num.java 1
multiplicationtable.java
nameprint.java
reversedigits.java
sumnaturalnum.java 1
swapnum.java 1

OUTLINE
TIMELINE
JAVA PROJECTS

```
      49: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
      52: ldc           #12                 // String Sum:
      54: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      57: iload         4
      59: invokevirtual #14                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      62: invokevirtual #15                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      65: invokevirtual #16                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      68: return
}
PS D:\CDAC\assignment> javap -v addtwonumbers
Classfile /D:/CDAC/assignment/addtwonumbers.class
  Last modified Feb 25, 2026; size 894 bytes
  MD5 checksum 0d6d8eb4fd874182216b2e18be06956c
  Compiled from "addtwonumbers.java"
public class addtwonumbers
  minor version: 0
  major version: 52
  flags: ACC_PUBLIC, ACC_SUPER
Constant pool:
   #1 = Methodref          #18.#27        // java/lang/Object."<init>":()V
   #2 = Class              #28            // java/util/Scanner
   #3 = Fieldref           #29.#30        // java/lang/System.in:Ljava/io/InputStream;
   #4 = Methodref          #2.#31         // java/util/Scanner."<init>":(Ljava/io/InputStream;)V
   #5 = Fieldref           #29.#32        // java/lang/System.out:Ljava/io/PrintStream;
   #6 = String             #33            // Enter the 1st number:
   #7 = Methodref          #34.#35        // java/io/PrintStream.print:(Ljava/lang/String;)V
   #8 = Methodref          #2.#36         // java/util/Scanner.nextInt:()I
   #9 = String             #37            // Enter the 2nd number:
  #10 = Class              #38            // java/lang/StringBuilder
  #11 = Methodref          #10.#27        // java/lang/StringBuilder."<init>":()V
  #12 = String             #39            // Sum:
  #13 = Methodref          #10.#40        // java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
  #14 = Methodref          #10.#41        // java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
  #15 = Methodref          #10.#42        // java/lang/StringBuilder.toString:()Ljava/lang/String;
  #16 = Methodref          #34.#43        // java/io/PrintStream.println:(Ljava/lang/String;)V
  #17 = Class              #44            // addtwonumbers
  #18 = Class              #45            // java/lang/Object
  #19 = Utf8               <init>
  #20 = Utf8               ()V
  #21 = Utf8               Code
  #22 = Utf8               LineNumberTable
```

factorialnum.java
fibonacciseries.java
largest3num.java 1
multiplicationtable.java
nameprint.java
reversedigits.java
sumnaturalnum.java 1
swapnum.java 1

```
  #62 = Utf8               println
{
  public addtwonumbers();
    descriptor: ()V
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=1, args_size=1
         0: aload_0
         1: invokespecial #1                  // Method java/lang/Object."<init>":()V
         4: return
      LineNumberTable:
        line 3: 0

  public static void main(java.lang.String[]);
    descriptor: ([Ljava/lang/String;)V
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=5, args_size=1
         0: new           #2                  // class java/util/Scanner
         3: dup
         4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
         7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
        10: astore_1
        11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        14: ldc           #6                  // String Enter the 1st number:
        16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
        19: aload_1
        20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
        23: istore_2
        24: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        27: ldc           #9                  // String Enter the 2nd number:
        29: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
        32: aload_1
        33: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
        36: istore_3
        37: iload_2
        38: iload_3
        39: iadd
        40: istore        4
        42: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        45: new           #10                 // class java/lang/StringBuilder
```

OUTLINE
TIMELINE
JAVA PROJECTS

sumnaturalnum.java 1
swapnum.java 1

```
        45: new           #10                 // class java/lang/StringBuilder
        48: dup
        49: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
        52: ldc           #12                 // String Sum:
        54: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
        57: iload         4
        59: invokevirtual #14                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
        62: invokevirtual #15                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
        65: invokevirtual #16                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
        68: return
      LineNumberTable:
        line 7: 0
        line 8: 11
        line 9: 19
        line 10: 24
        line 11: 32
        line 12: 37
        line 13: 42
        line 14: 68
}
SourceFile: "addtwonumbers.java"
PS D:\CDAC\assignment>
```

factorialnum.java
fibonacciseries.java
largest3num.java 1
multiplicationtable.java
nameprint.java
reversedigits.java
sumnaturalnum.java 1
swapnum.java 1

```
PS D:\CDAC\assignment> javap -v addtwonumbers
  #22 = Utf8               LineNumberTable
  #23 = Utf8               main
  #24 = Utf8               ([Ljava/lang/String;)V
  #25 = Utf8               SourceFile
  #26 = Utf8               addtwonumbers.java
  #27 = NameAndType        #19:#20       // "<init>":()V
  #28 = Utf8               java/util/Scanner
  #29 = Class              #46           // java/lang/System
  #30 = NameAndType        #47:#48       // in:Ljava/io/InputStream;
  #31 = NameAndType        #19:#49       // "<init>":(Ljava/io/InputStream;)V
  #32 = NameAndType        #50:#51       // out:Ljava/io/PrintStream;
  #33 = Utf8               Enter the 1st number:
  #34 = Class              #52           // java/io/PrintStream
  #35 = NameAndType        #53:#54       // print:(Ljava/lang/String;)V
  #36 = NameAndType        #55:#56       // nextInt:()I
  #37 = Utf8               Enter the 2nd number:
  #38 = Utf8               java/lang/StringBuilder
  #39 = Utf8               Sum:
  #40 = NameAndType        #57:#58       // append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
  #41 = NameAndType        #57:#59       // append:(I)Ljava/lang/StringBuilder;
  #42 = NameAndType        #60:#61       // toString:()Ljava/lang/String;
  #43 = NameAndType        #62:#54       // println:(Ljava/lang/String;)V
  #44 = Utf8               addtwonumbers
  #45 = Utf8               java/lang/Object
  #46 = Utf8               java/lang/System
  #47 = Utf8               in
  #48 = Utf8               Ljava/io/InputStream;
  #49 = Utf8               (Ljava/io/InputStream;)V
  #50 = Utf8               out
  #51 = Utf8               Ljava/io/PrintStream;
  #52 = Utf8               java/io/PrintStream
  #53 = Utf8               print
  #54 = Utf8               (Ljava/lang/String;)V
  #55 = Utf8               nextInt
  #56 = Utf8               ()I
  #57 = Utf8               append
  #58 = Utf8               (Ljava/lang/String;)Ljava/lang/StringBuilder;
  #59 = Utf8               (I)Ljava/lang/StringBuilder;
  #60 = Utf8               toString
  #61 = Utf8               ()Ljava/lang/String;
  #62 = Utf8               println
```

OUTLINE
TIMELINE
JAVA PROJECTS

Question 2:

EXPLORER

ASSIGNMENT

- addtwonumbers.class
- addtwonumbers.java 1
- evenorodd.java
- factorialnum.java
- fibonacciseries.class
- fibonacciseries.java
- largest3num.java 1
- multiplicationtable.java
- nameprint.java
- reversedigits.java
- sumnaturalnum.java 1
- swapnum.java 1

OUTLINE

TIMELINE

JAVA PROJECTS

fibonacciseries.java

fibonacciseries.java > fibonacciseries

```java
public class fibonacciseries {

    Run | Debug
    public static void main(String[] args) {


        int num1 = 0;
        int num2 = 1;

        for (int i = 1; i <= 10; i++) {

            System.out.print(num1 + " ");

            int num3 = num1 + num2;
            num1 = num2;
            num2 = num3;
        }
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CDAC\assignment> javac ./fibonacciseries.java
PS D:\CDAC\assignment> java fibonacciseries
0 1 1 2 3 5 8 13 21 34
PS D:\CDAC\assignment> javap fibonacciseries
Compiled from "fibonacciseries.java"
public class fibonacciseries {
  public fibonacciseries();
  public static void main(java.lang.String[]);
}
PS D:\CDAC\assignment> javap -c fibonacciseries
Compiled from "fibonacciseries.java"
public class fibonacciseries {
  public fibonacciseries();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: iconst_0
       1: istore_1
```

```
PS D:\CDAC\assignment> javap -c fibonacciseries
       5: istore_3
       6: iload_3
       7: bipush        10
       9: if_icmpgt     53
      12: getstatic     #2                  // Field java/lang/System.out:Ljava/io/PrintStream;
      15: new           #3                  // class java/lang/StringBuilder
      18: dup
      19: invokespecial #4                  // Method java/lang/StringBuilder."<init>":()V
      22: iload_1
      23: invokevirtual #5                  // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      26: ldc           #6                  // String
      28: invokevirtual #7                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      31: invokevirtual #8                  // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      34: invokevirtual #9                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
      37: iload_1
      38: iload_2
      39: iadd
      40: istore        4
      42: iload_2
      43: istore_1
      44: iload         4
      46: istore_2
      47: iinc          3, 1
      50: goto          6
      53: return
}
PS D:\CDAC\assignment> javap -v fibonacciseries
Classfile /D:/CDAC/assignment/fibonacciseries.class
  Last modified Feb 25, 2026; size 715 bytes
  MD5 checksum 6ea8d04cfe01dd27d1a9c528d17b694b
  Compiled from "fibonacciseries.java"
public class fibonacciseries
  minor version: 0
  major version: 52
  flags: ACC_PUBLIC, ACC_SUPER
Constant pool:
   #1 = Methodref          #11.#21        // java/lang/Object."<init>":()V
   #2 = Fieldref           #22.#23        // java/lang/System.out:Ljava/io/PrintStream;
   #3 = Class              #24            // java/lang/StringBuilder
   #4 = Methodref          #3.#21         // java/lang/StringBuilder."<init>":()V
   #5 = Methodref          #3.#25         // java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
   #6 = String             #26            //
   #7 = Methodref          #3.#27         // java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
   #8 = Methodref          #3.#28         // java/lang/StringBuilder.toString:()Ljava/lang/String;
   #9 = Methodref          #29.#30        // java/io/PrintStream.print:(Ljava/lang/String;)V
```

```
PS D:\CDAC\assignment> javap -v fibonacciseries
  public static void main(java.lang.String[]);
    descriptor: ([Ljava/lang/String;)V
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=5, args_size=1
         0: iconst_0
         1: istore_1
         2: iconst_1
         3: istore_2
         4: iconst_1
         5: istore_3
         6: iload_3
         7: bipush        10
         9: if_icmpgt     53
        12: getstatic     #2                  // Field java/lang/System.out:Ljava/io/PrintStream;
        15: new           #3                  // class java/lang/StringBuilder
        18: dup
        19: invokespecial #4                  // Method java/lang/StringBuilder."<init>":()V
        22: iload_1
        23: invokevirtual #5                  // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
        26: ldc           #6                  // String
        28: invokevirtual #7                  // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
        31: invokevirtual #8                  // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
        34: invokevirtual #9                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
        37: iload_1
        38: iload_2
        39: iadd
        40: istore        4
        42: iload_2
        43: istore_1
        44: iload         4
        46: istore_2
        47: iinc          3, 1
        50: goto          6
        53: return
      LineNumberTable:
        line 6: 0
        line 7: 2
        line 9: 4
        line 11: 12
        line 13: 37
        line 14: 42
        line 15: 44
        line 9: 47
        line 17: 53
```

```
PS D:\CDAC\assignment> javap -v fibonacciseries
  #10 = Class              #31            // fibonacciseries
  #11 = Class              #32            // java/lang/Object
  #12 = Utf8               <init>
  #13 = Utf8               ()V
  #14 = Utf8               Code
  #15 = Utf8               LineNumberTable
  #16 = Utf8               main
  #17 = Utf8               ([Ljava/lang/String;)V
  #18 = Utf8               StackMapTable
  #19 = Utf8               SourceFile
  #20 = Utf8               fibonacciseries.java
  #21 = NameAndType        #12:#13        // "<init>":()V
  #22 = Class              #33            // java/lang/System
  #23 = NameAndType        #34:#35        // out:Ljava/io/PrintStream;
  #24 = Utf8               java/lang/StringBuilder
  #25 = NameAndType        #36:#37        // append:(I)Ljava/lang/StringBuilder;
  #26 = Utf8
  #27 = NameAndType        #36:#38        // append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
  #28 = NameAndType        #39:#40        // toString:()Ljava/lang/String;
  #29 = Class              #41            // java/io/PrintStream
  #30 = NameAndType        #42:#43        // print:(Ljava/lang/String;)V
  #31 = Utf8               fibonacciseries
  #32 = Utf8               java/lang/Object
  #33 = Utf8               java/lang/System
  #34 = Utf8               out
  #35 = Utf8               Ljava/io/PrintStream;
  #36 = Utf8               append
  #37 = Utf8               (I)Ljava/lang/StringBuilder;
  #38 = Utf8               (Ljava/lang/String;)Ljava/lang/StringBuilder;
  #39 = Utf8               toString
  #40 = Utf8               ()Ljava/lang/String;
  #41 = Utf8               java/io/PrintStream
  #42 = Utf8               print
  #43 = Utf8               (Ljava/lang/String;)V
{
  public fibonacciseries();
    descriptor: ()V
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=1, args_size=1
         0: aload_0
         1: invokespecial #1                  // Method java/lang/Object."<init>":()V
         4: return
      LineNumberTable:
        line 1: 0
```

```
PS D:\CDAC\assignment> javap -v fibonacciseries
         31: invokevirtual #8                  // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
         34: invokevirtual #9                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
         37: iload_1
         38: iload_2
         39: iadd
         40: istore        4
         42: iload_2
         43: istore_1
         44: iload         4
         46: istore_2
         47: iinc          3, 1
         50: goto          6
         53: return
       LineNumberTable:
         line 6: 0
         line 7: 2
         line 9: 4
         line 11: 12
         line 13: 37
         line 14: 42
         line 15: 44
         line 9: 47
         line 17: 53
       StackMapTable: number_of_entries = 2
         frame_type = 254 /* append */
           offset_delta = 6
           locals = [ int, int, int ]
         frame_type = 250 /* chop */
           offset_delta = 46
}
SourceFile: "fibonacciseries.java"
PS D:\CDAC\assignment>
```

Question 3:

ASSIGNMENT

- addtwonumbers.class
- addtwonumbers.java 1
- evenorodd.java
- factorialnum.java
- fibonacciseries.class
- fibonacciseries.java
- largest3num.java 1
- multiplicationtable.java
- nameprint.class
- nameprint.java
- reversedigits.java
- sumnaturalnum.java 1
- swapnum.java 1

OUTLINE
TIMELINE
JAVA PROJECTS

nameprint.java > nameprint > main(String[])

```java
public class nameprint {

    Run | Debug
    public static void main(String[] args) {


        for (int i = 1; i <= 10; i++) {

            System.out.println(x: "Priyansu Pattanaik");

        }

    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CDAC\assignment> javac ./nameprint.java
PS D:\CDAC\assignment> java nameprint
Priyansu Pattanaik
Priyansu Pattanaik
Priyansu Pattanaik
Priyansu Pattanaik
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Priyansu Pattanaik
PS D:\CDAC\assignment> javap nameprint
Compiled from "nameprint.java"
public class nameprint {
  public nameprint();
```

```
PS D:\CDAC\assignment> javap nameprint
Compiled from "nameprint.java"
public class nameprint {
  public nameprint();
  public static void main(java.lang.String[]);
}
PS D:\CDAC\assignment> javap -c nameprint
Compiled from "nameprint.java"
public class nameprint {
  public nameprint();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: iconst_1
       1: istore_1
       2: iload_1
       3: bipush        10
       5: if_icmpgt     22
       8: getstatic     #2                  // Field java/lang/System.out:Ljava/io/PrintStream;
      11: ldc           #3                  // String Priyansu Pattanaik
      13: invokevirtual #4                  // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      16: iinc          1, 1
      19: goto          2
      22: return
}
```

```
PS D:\CDAC\assignment> javap -c nameprint
PS D:\CDAC\assignment> javap -v nameprint
Classfile /D:/CDAC/assignment/nameprint.class
  Last modified Feb 25, 2026; size 483 bytes
  MD5 checksum a411516638ce0c5a067087e0ee74b3bf
  Compiled from "nameprint.java"
public class nameprint
  minor version: 0
  major version: 52
  flags: ACC_PUBLIC, ACC_SUPER
Constant pool:
   #1 = Methodref          #6.#16         // java/lang/Object."<init>":()V
   #2 = Fieldref           #17.#18        // java/lang/System.out:Ljava/io/PrintStream;
   #3 = String             #19            // Priyansu Pattanaik
   #4 = Methodref          #20.#21        // java/io/PrintStream.println:(Ljava/lang/String;)V
   #5 = Class              #22            // nameprint
   #6 = Class              #23            // java/lang/Object
   #7 = Utf8               <init>
   #8 = Utf8               ()V
   #9 = Utf8               Code
  #10 = Utf8               LineNumberTable
  #11 = Utf8               main
  #12 = Utf8               ([Ljava/lang/String;)V
  #13 = Utf8               StackMapTable
  #14 = Utf8               SourceFile
  #15 = Utf8               nameprint.java
  #16 = NameAndType        #7:#8          // "<init>":()V
  #17 = Class              #24            // java/lang/System
  #18 = NameAndType        #25:#26        // out:Ljava/io/PrintStream;
  #19 = Utf8               Priyansu Pattanaik
  #20 = Class              #27            // java/io/PrintStream
  #21 = NameAndType        #28:#29        // println:(Ljava/lang/String;)V
  #22 = Utf8               nameprint
  #23 = Utf8               java/lang/Object
  #24 = Utf8               java/lang/System
  #25 = Utf8               out
  #26 = Utf8               Ljava/io/PrintStream;
  #27 = Utf8               java/io/PrintStream
  #28 = Utf8               println
  #29 = Utf8               (Ljava/lang/String;)V
{
  public nameprint();
```

```
PS D:\CDAC\assignment> javap -v nameprint
{
  public nameprint();
    descriptor: ()V
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=1, args_size=1
         0: aload_0
         1: invokespecial #1                  // Method java/lang/Object."<init>":()V
         4: return
      LineNumberTable:
        line 1: 0

  public static void main(java.lang.String[]);
    descriptor: ([Ljava/lang/String;)V
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=2, locals=2, args_size=1
         0: iconst_1
         1: istore_1
         2: iload_1
         3: bipush        10
         5: if_icmpgt     22
         8: getstatic     #2                  // Field java/lang/System.out:Ljava/io/PrintStream;
        11: ldc           #3                  // String Priyansu Pattanaik
        13: invokevirtual #4                  // Method java/io/PrintStream.println:(Ljava/lang/String;)V
        16: iinc          1, 1
        19: goto          2
        22: return
      LineNumberTable:
        line 6: 0
        line 8: 8
        line 6: 16
        line 12: 22
      StackMapTable: number_of_entries = 2
        frame_type = 252 /* append */
          offset_delta = 2
          locals = [ int ]
        frame_type = 250 /* chop */
          offset_delta = 19
}
SourceFile: "nameprint.java"
```

Question 4 :

EXPLORER

ASSIGNMENT

- addtwonumbers.class
- addtwonumbers.java 1
- evenorodd.java
- factorialnum.class
- factorialnum.java
- fibonacciseries.class
- fibonacciseries.java
- largest3num.java 1
- multiplicationtable.java
- nameprint.class
- nameprint.java
- reversedigits.java
- sumnaturalnum.java 1
- swapnum.java 1

OUTLINE

TIMELINE

JAVA PROJECTS

factorialnum.java

factorialnum.java > factorialnum > main(String[])

```java
public class factorialnum {

    Run | Debug
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);
        System.out.print(s: "Enter num: ");
        int num = input.nextInt();
        int factorial = 1;
        for (int i = 1; i <= num; i++) {
            factorial = factorial * i;
        }
        System.out.println("Factorial of " + num + ": " + factorial);

        input.close();
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CDAC\assignment> javac ./factorialnum.java
PS D:\CDAC\assignment> java factorialnum
Enter num: 5
Factorial of 5: 120
PS D:\CDAC\assignment> javap factorialnum
Compiled from "factorialnum.java"
public class factorialnum {
  public factorialnum();
  public static void main(java.lang.String[]);
}
PS D:\CDAC\assignment> javap -c factorialnum
Compiled from "factorialnum.java"
public class factorialnum {
  public factorialnum();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
```

```
PS D:\CDAC\assignment> javap -c factorialnum
    Code:
       0: new           #2                  // class java/util/Scanner
       3: dup
       4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
       7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
      10: astore_1
      11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      14: ldc           #6                  // String Enter num:
      16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
      19: aload_1
      20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
      23: istore_2
      24: iconst_1
      25: istore_3
      26: iconst_1
      27: istore        4
      29: iload         4
      31: iload_2
      32: if_icmpgt     46
      35: iload_3
      36: iload         4
      38: imul
      39: istore_3
      40: iinc          4, 1
      43: goto          29
      46: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      49: new           #9                  // class java/lang/StringBuilder
      52: dup
      53: invokespecial #10                 // Method java/lang/StringBuilder."<init>":()V
      56: ldc           #11                 // String Factorial of
      58: invokevirtual #12                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      61: iload_2
      62: invokevirtual #13                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      65: ldc           #14                 // String :
      67: invokevirtual #12                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      70: iload_3
      71: invokevirtual #13                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      74: invokevirtual #15                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      77: invokevirtual #16                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      80: aload_1
      81: invokevirtual #17                 // Method java/util/Scanner.close:()V
```

```
PS D:\CDAC\assignment> javap -c factorialnum
PS D:\CDAC\assignment> javap -v factorialnum
Classfile /D:/CDAC/assignment/factorialnum.class
  Last modified Feb 25, 2026; size 982 bytes
  MD5 checksum 53c3ef77b2d5b545333e3b06f2729dbe
  Compiled from "factorialnum.java"
public class factorialnum
  minor version: 0
  major version: 52
  flags: ACC_PUBLIC, ACC_SUPER
Constant pool:
   #1 = Methodref          #19.#31        // java/lang/Object."<init>":()V
   #2 = Class              #32            // java/util/Scanner
   #3 = Fieldref           #33.#34        // java/lang/System.in:Ljava/io/InputStream;
   #4 = Methodref          #2.#35         // java/util/Scanner."<init>":(Ljava/io/InputStream;)V
   #5 = Fieldref           #33.#36        // java/lang/System.out:Ljava/io/PrintStream;
   #6 = String             #37            // Enter num:
   #7 = Methodref          #38.#39        // java/io/PrintStream.print:(Ljava/lang/String;)V
   #8 = Methodref          #2.#40         // java/util/Scanner.nextInt:()I
   #9 = Class              #41            // java/lang/StringBuilder
  #10 = Methodref          #9.#31         // java/lang/StringBuilder."<init>":()V
  #11 = String             #42            // Factorial of
  #12 = Methodref          #9.#43         // java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
  #13 = Methodref          #9.#44         // java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
  #14 = String             #45            // :
  #15 = Methodref          #9.#46         // java/lang/StringBuilder.toString:()Ljava/lang/String;
  #16 = Methodref          #38.#47        // java/io/PrintStream.println:(Ljava/lang/String;)V
  #17 = Methodref          #2.#48         // java/util/Scanner.close:()V
  #18 = Class              #49            // factorialnum
  #19 = Class              #50            // java/lang/Object
  #20 = Utf8               <init>
  #21 = Utf8               ()V
  #22 = Utf8               Code
  #23 = Utf8               LineNumberTable
  #24 = Utf8               main
  #25 = Utf8               ([Ljava/lang/String;)V
  #26 = Utf8               StackMapTable
  #27 = Class              #51            // "[Ljava/lang/String;"
  #28 = Class              #32            // java/util/Scanner
  #29 = Utf8               SourceFile
  #30 = Utf8               factorialnum.java
  #31 = NameAndType        #20:#21        // "<init>":()V
```

```
PS D:\CDAC\assignment> javap -v factorialnum
  #30 = Utf8               factorialnum.java
  #31 = NameAndType        #20:#21        // "<init>":()V
  #32 = Utf8               java/util/Scanner
  #33 = Class              #52            // java/lang/System
  #34 = NameAndType        #53:#54        // in:Ljava/io/InputStream;
  #35 = NameAndType        #20:#55        // "<init>":(Ljava/io/InputStream;)V
  #36 = NameAndType        #56:#57        // out:Ljava/io/PrintStream;
  #37 = Utf8               Enter num:
  #38 = Class              #58            // java/io/PrintStream
  #39 = NameAndType        #59:#60        // print:(Ljava/lang/String;)V
  #40 = NameAndType        #61:#62        // nextInt:()I
  #41 = Utf8               java/lang/StringBuilder
  #42 = Utf8               Factorial of
  #43 = NameAndType        #63:#64        // append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
  #44 = NameAndType        #63:#65        // append:(I)Ljava/lang/StringBuilder;
  #45 = Utf8               :
  #46 = NameAndType        #66:#67        // toString:()Ljava/lang/String;
  #47 = NameAndType        #68:#60        // println:(Ljava/lang/String;)V
  #48 = NameAndType        #69:#21        // close:()V
  #49 = Utf8               factorialnum
  #50 = Utf8               java/lang/Object
  #51 = Utf8               [Ljava/lang/String;
  #52 = Utf8               java/lang/System
  #53 = Utf8               in
  #54 = Utf8               Ljava/io/InputStream;
  #55 = Utf8               (Ljava/io/InputStream;)V
  #56 = Utf8               out
  #57 = Utf8               Ljava/io/PrintStream;
  #58 = Utf8               java/io/PrintStream
  #59 = Utf8               print
  #60 = Utf8               (Ljava/lang/String;)V
  #61 = Utf8               nextInt
  #62 = Utf8               ()I
  #63 = Utf8               append
  #64 = Utf8               (Ljava/lang/String;)Ljava/lang/StringBuilder;
  #65 = Utf8               (I)Ljava/lang/StringBuilder;
  #66 = Utf8               toString
  #67 = Utf8               ()Ljava/lang/String;
  #68 = Utf8               println
  #69 = Utf8               close
{
```

```
PS D:\CDAC\assignment> javap -v factorialnum
{
  public factorialnum();
    descriptor: ()V
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=1, args_size=1
         0: aload_0
         1: invokespecial #1                  // Method java/lang/Object."<init>":()V
         4: return
      LineNumberTable:
        line 3: 0

  public static void main(java.lang.String[]);
    descriptor: ([Ljava/lang/String;)V
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=5, args_size=1
         0: new           #2                  // class java/util/Scanner
         3: dup
         4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
         7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
        10: astore_1
        11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        14: ldc           #6                  // String Enter num:
        16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
        19: aload_1
        20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
        23: istore_2
        24: iconst_1
        25: istore_3
        26: iconst_1
        27: istore        4
        29: iload         4
        31: iload_2
        32: if_icmpgt     46
        35: iload_3
        36: iload         4
        38: imul
        39: istore_3
        40: iinc          4, 1
        43: goto          29
```

```
PS D:\CDAC\assignment> javap -v factorialnum
         65: ldc           #14                 // String :
         65: ldc           #14                 // String :
         67: invokevirtual #12                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
         67: invokevirtual #12                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
         70: iload_3
         70: iload_3
         71: invokevirtual #13                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
         74: invokevirtual #15                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
         71: invokevirtual #13                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
         74: invokevirtual #15                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
         77: invokevirtual #16                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
         80: aload_1
         77: invokevirtual #16                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
         80: aload_1
         80: aload_1
         81: invokevirtual #17                 // Method java/util/Scanner.close:()V
         81: invokevirtual #17                 // Method java/util/Scanner.close:()V
         84: return
       LineNumberTable:
         line 7: 0
         line 8: 11
         line 9: 19
         line 10: 24
         line 11: 26
         line 12: 35
         line 11: 40
         line 14: 46
         line 16: 80
         line 17: 84
       StackMapTable: number_of_entries = 2
         frame_type = 255 /* full_frame */
           offset_delta = 29
           locals = [ class "[Ljava/lang/String;", class java/util/Scanner, int, int, int ]
           stack = []
         frame_type = 250 /* chop */
           offset_delta = 16
}
SourceFile: "factorialnum.java"
PS D:\CDAC\assignment>
```

Question 5:

ASSIGNMENT

- addtwonumbers.class
- addtwonumbers.java 1
- evenorodd.class
- evenorodd.java
- factorialnum.class
- factorialnum.java
- fibonacciseries.class
- fibonacciseries.java
- largest3num.java 1
- multiplicationtable.java
- nameprint.class
- nameprint.java
- reversedigits.java
- sumnaturalnum.java 1
- swapnum.java 1

OUTLINE
TIMELINE
JAVA PROJECTS

evenorodd.java > evenorodd

```java
import java.util.Scanner;

public class evenorodd {

    Run | Debug
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print(s: "Enter number: ");
        int num = input.nextInt();
        if (num % 2 == 0) {
            System.out.println(num + " Even number.");
        } else {
            System.out.println(num + " Odd number.");
        }
        input.close();
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

```
PS D:\CDAC\assignment> javac ./evenorodd.java
PS D:\CDAC\assignment> java evenorodd
Enter number: 13
13 Odd number.
PS D:\CDAC\assignment> javap evenorodd
Compiled from "evenorodd.java"
public class evenorodd {
  public evenorodd();
  public static void main(java.lang.String[]);
}
PS D:\CDAC\assignment> javap -c evenorodd
Compiled from "evenorodd.java"
public class evenorodd {
  public evenorodd();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: new           #2                  // class java/util/Scanner
```

```
PS D:\CDAC\assignment> javap -c evenorodd

  public static void main(java.lang.String[]);
    Code:
       0: new           #2                  // class java/util/Scanner
       3: dup
       4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
       7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
      10: astore_1
      11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      14: ldc           #6                  // String Enter number:
      16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
      19: aload_1
      20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
      23: istore_2
      24: iload_2
      25: iconst_2
      26: irem
      27: ifne          58
      30: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      33: new           #9                  // class java/lang/StringBuilder
      36: dup
      37: invokespecial #10                 // Method java/lang/StringBuilder."<init>":()V
      40: iload_2
      41: invokevirtual #11                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      44: ldc           #12                 // String  Even number.
      46: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      49: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      52: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      55: goto          83
      58: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      61: new           #9                  // class java/lang/StringBuilder
      64: dup
      65: invokespecial #10                 // Method java/lang/StringBuilder."<init>":()V
      68: iload_2
      69: invokevirtual #11                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      72: ldc           #16                 // String  Odd number.
      74: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
```

```
PS D:\CDAC\assignment> javac -v evenorodd
PS D:\CDAC\assignment> javap -v evenorodd
Classfile /D:/CDAC/assignment/evenorodd.class
  Last modified Feb 25, 2026; size 948 bytes
  MD5 checksum c83a49dfcf2f64600e356061d6a30673
  Compiled from "evenorodd.java"
public class evenorodd
  minor version: 0
  major version: 52
  flags: ACC_PUBLIC, ACC_SUPER
Constant pool:
   #1 = Methodref          #19.#30        // java/lang/Object."<init>":()V
   #2 = Class              #31            // java/util/Scanner
   #3 = Fieldref           #32.#33        // java/lang/System.in:Ljava/io/InputStream;
   #4 = Methodref          #2.#34         // java/util/Scanner."<init>":(Ljava/io/InputStream;)V
   #5 = Fieldref           #32.#35        // java/lang/System.out:Ljava/io/PrintStream;
   #6 = String             #36            // Enter number:
   #7 = Methodref          #37.#38        // java/io/PrintStream.print:(Ljava/lang/String;)V
   #8 = Methodref          #2.#39         // java/util/Scanner.nextInt:()I
   #9 = Class              #40            // java/lang/StringBuilder
  #10 = Methodref          #9.#30         // java/lang/StringBuilder."<init>":()V
  #11 = Methodref          #9.#41         // java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
  #12 = String             #42            //  Even number.
  #13 = Methodref          #9.#43         // java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
  #14 = Methodref          #9.#44         // java/lang/StringBuilder.toString:()Ljava/lang/String;
  #15 = Methodref          #37.#45        // java/io/PrintStream.println:(Ljava/lang/String;)V
  #16 = String             #46            //  Odd number.
  #17 = Methodref          #2.#47         // java/util/Scanner.close:()V
  #18 = Class              #48            // evenorodd
  #19 = Class              #49            // java/lang/Object
  #20 = Utf8               <init>
  #21 = Utf8               ()V
  #22 = Utf8               Code
  #23 = Utf8               LineNumberTable
  #24 = Utf8               main
  #25 = Utf8               ([Ljava/lang/String;)V
  #26 = Utf8               StackMapTable
  #27 = Class              #31            // java/util/Scanner
```

```
PS D:\CDAC\assignment> javap -v evenorodd
  #28 = Utf8               SourceFile
  #29 = Utf8               evenorodd.java
  #30 = NameAndType        #20:#21        // "<init>":()V
  #31 = Utf8               java/util/Scanner
  #32 = Class              #50            // java/lang/System
  #33 = NameAndType        #51:#52        // in:Ljava/io/InputStream;
  #34 = NameAndType        #20:#53        // "<init>":(Ljava/io/InputStream;)V
  #35 = NameAndType        #54:#55        // out:Ljava/io/PrintStream;
  #36 = Utf8               Enter number:
  #37 = Class              #56            // java/io/PrintStream
  #38 = NameAndType        #57:#58        // print:(Ljava/lang/String;)V
  #39 = NameAndType        #59:#60        // nextInt:()I
  #40 = Utf8               java/lang/StringBuilder
  #41 = NameAndType        #61:#62        // append:(I)Ljava/lang/StringBuilder;
  #42 = Utf8                Even number.
  #43 = NameAndType        #61:#63        // append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
  #44 = NameAndType        #64:#65        // toString:()Ljava/lang/String;
  #45 = NameAndType        #66:#58        // println:(Ljava/lang/String;)V
  #46 = Utf8                Odd number.
  #47 = NameAndType        #67:#21        // close:()V
  #48 = Utf8               evenorodd
  #49 = Utf8               java/lang/Object
  #50 = Utf8               java/lang/System
  #51 = Utf8               in
  #52 = Utf8               Ljava/io/InputStream;
  #53 = Utf8               (Ljava/io/InputStream;)V
  #54 = Utf8               out
  #55 = Utf8               Ljava/io/PrintStream;
  #56 = Utf8               java/io/PrintStream
  #57 = Utf8               print
  #58 = Utf8               (Ljava/lang/String;)V
  #59 = Utf8               nextInt
  #60 = Utf8               ()I
  #61 = Utf8               append
  #62 = Utf8               (I)Ljava/lang/StringBuilder;
  #63 = Utf8               (Ljava/lang/String;)Ljava/lang/StringBuilder;
  #64 = Utf8               toString
```

```
PS D:\CDAC\assignment> javap -v evenorodd
{
  public evenorodd();
    descriptor: ()V
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=1, args_size=1
         0: aload_0
         1: invokespecial #1                  // Method java/lang/Object."<init>":()V
         4: return
      LineNumberTable:
        line 3: 0

  public static void main(java.lang.String[]);
    descriptor: ([Ljava/lang/String;)V
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=3, args_size=1
         0: new           #2                  // class java/util/Scanner
         3: dup
         4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
         7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
        10: astore_1
        11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        14: ldc           #6                  // String Enter number:
        16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
        19: aload_1
        20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
        23: istore_2
        24: iload_2
        25: iconst_2
        26: irem
        27: ifne          58
        30: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        33: new           #9                  // class java/lang/StringBuilder
        36: dup
        37: invokespecial #10                 // Method java/lang/StringBuilder."<init>":()V
        40: iload_2
```

```
PS D:\CDAC\assignment> javap -v evenorodd
      49: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      52: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      55: goto          83
      58: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      61: new           #9                  // class java/lang/StringBuilder
      64: dup
      33: new           #9                  // class java/lang/StringBuilder
      36: dup
      37: invokespecial #10                 // Method java/lang/StringBuilder."<init>":()V
      40: iload_2
      41: invokevirtual #11                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      44: ldc           #12                 // String  Even number.
      46: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      49: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      52: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      55: goto          83
      58: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      61: new           #9                  // class java/lang/StringBuilder
      64: dup
      37: invokespecial #10                 // Method java/lang/StringBuilder."<init>":()V
      40: iload_2
      41: invokevirtual #11                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      44: ldc           #12                 // String  Even number.
      46: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      49: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      52: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      55: goto          83
      58: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      61: new           #9                  // class java/lang/StringBuilder
      64: dup
      41: invokevirtual #11                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      44: ldc           #12                 // String  Even number.
      46: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      49: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      52: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      55: goto          83
      58: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
```

```
      49: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      52: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      55: goto          83
      58: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      61: new           #9                  // class java/lang/StringBuilder
      64: dup
      55: goto          83
      58: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      61: new           #9                  // class java/lang/StringBuilder
      64: dup
      61: new           #9                  // class java/lang/StringBuilder
      64: dup
      64: dup
      65: invokespecial #10                 // Method java/lang/StringBuilder."<init>":()V
      65: invokespecial #10                 // Method java/lang/StringBuilder."<init>":()V
      68: iload_2
      69: invokevirtual #11                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      72: ldc           #16                 // String  Odd number.
      74: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      77: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      80: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      83: aload_1
      84: invokevirtual #17                 // Method java/util/Scanner.close:()V
      87: return
    LineNumberTable:
      line 6: 0
      line 7: 11
      line 8: 19
      line 9: 24
      line 10: 30
      line 12: 58
      line 14: 83
      line 15: 87
    StackMapTable: number_of_entries = 2
      frame_type = 253 /* append */
        offset_delta = 58
        locals = [ class java/util/Scanner, int ]
      frame_type = 24 /* same */
```


```
      72: ldc           #16                 // String  Odd number.
      74: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      77: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      80: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      83: aload_1
      84: invokevirtual #17                 // Method java/util/Scanner.close:()V
      87: return
    LineNumberTable:
      line 6: 0
      line 7: 11
      line 8: 19
      line 9: 24
      line 10: 30
      line 12: 58
      line 14: 83
      line 15: 87
    StackMapTable: number_of_entries = 2
      frame_type = 253 /* append */
        offset_delta = 58
        locals = [ class java/util/Scanner, int ]
      frame_type = 24 /* same */
```


```
}
SourceFile: "evenorodd.java"
PS D:\CDAC\assignment>
```

Question 6:

EXPLORER

ASSIGNMENT

- addtwonumbers.class
- addtwonumbers.java 1
- evenorodd.class
- evenorodd.java
- factorialnum.class
- factorialnum.java
- fibonacciseries.class
- fibonacciseries.java
- largest3num.class
- largest3num.java 1
- multiplicationtable.java
- nameprint.class
- nameprint.java
- reversedigits.java
- sumnaturalnum.java 1
- swapnum.java 1

OUTLINE

TIMELINE

JAVA PROJECTS

largest3num.java 1

largest3num.java > largest3num > main(String[])

```java
import java.util.Scanner;

public class largest3num {
    Run | Debug
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);
        System.out.print(s: "Enter num1: ");
        int num1 = input.nextInt();
        System.out.print(s: "Enter num2: ");
        int num2 = input.nextInt();
        System.out.print(s: "Enter num3: ");
        int num3 = input.nextInt();

        if (num1 > num2 && num1 > num3) {
            System.out.println("The largest num is: " + num1);
        } else if (num2 > num1 && num2 > num3) {
            System.out.println("The largest num is: " + num2);
        } else if (num3 > num2 && num3 > num1) {
            System.out.println("The largest num is: " + num3);
        }
        else if (num1==num2 && num1==num3){
            System.out.println(x: "all numbers equal");
        }
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

```
PS D:\CDAC\assignment>
```

```
PS D:\CDAC\assignment>
PS D:\CDAC\assignment> javac ./largest3num.java
PS D:\CDAC\assignment> java largest3num
Enter num1: 12
Enter num2: 14
Enter num3: 15
The largest num is: 15
PS D:\CDAC\assignment> javap largest3num
Compiled from "largest3num.java"
public class largest3num {
```

```
PS D:\CDAC\assignment> javap largest3num
PS D:\CDAC\assignment> javap -c largest3num
Compiled from "largest3num.java"
public class largest3num {
  public largest3num();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: new           #2                  // class java/util/Scanner
       3: dup
       4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
       7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
      10: astore_1
      11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      14: ldc           #6                  // String Enter num1:
      16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
      19: aload_1
      20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
      23: istore_2
      24: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      27: ldc           #9                  // String Enter num2:
      29: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
      32: aload_1
      33: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
      36: istore_3
      37: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      40: ldc           #10                 // String Enter num3:
      42: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
      45: aload_1
      46: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
      49: istore        4
      51: iload_2
      52: iload_3
      53: if_icmple     90
      56: iload_2
      57: iload         4
      59: if_icmple     90
      62: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
```

```
PS D:\CDAC\assignment> javap -c largest3num
      59: if_icmple     90
      62: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      65: new           #11                 // class java/lang/StringBuilder
      68: dup
      69: invokespecial #12                 // Method java/lang/StringBuilder."<init>":()V
      72: ldc           #13                 // String The largest num is:
      74: invokevirtual #14                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      77: iload_2
      78: invokevirtual #15                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      81: invokevirtual #16                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      84: invokevirtual #17                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      87: goto          189
      90: iload_3
      91: iload_2
      92: if_icmple     129
      95: iload_3
      96: iload         4
      98: if_icmple     129
     101: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
     104: new           #11                 // class java/lang/StringBuilder
     107: dup
     108: invokespecial #12                 // Method java/lang/StringBuilder."<init>":()V
     111: ldc           #13                 // String The largest num is:
     113: invokevirtual #14                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
     116: iload_3
     117: invokevirtual #15                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
     120: invokevirtual #16                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
     123: invokevirtual #17                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
     126: goto          189
     129: iload         4
     131: iload_3
     132: if_icmple     170
     135: iload         4
     137: iload_2
     138: if_icmple     170
     141: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
     144: new           #11                 // class java/lang/StringBuilder
     147: dup
     148: invokespecial #12                 // Method java/lang/StringBuilder."<init>":()V
     151: ldc           #13                 // String The largest num is:
     153: invokevirtual #14                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
```

```
PS D:\CDAC\assignment> javap -c largest3num
PS D:\CDAC\assignment> javap -v largest3num
Classfile /D:/CDAC/assignment/largest3num.class
  Last modified Feb 25, 2026; size 1150 bytes
  MD5 checksum cc150f0501b2df95dc8d99b2b21cc44f
  Compiled from "largest3num.java"
public class largest3num
  minor version: 0
  major version: 52
  flags: ACC_PUBLIC, ACC_SUPER
Constant pool:
   #1 = Methodref          #20.#32        // java/lang/Object."<init>":()V
   #2 = Class              #33            // java/util/Scanner
   #3 = Fieldref           #34.#35        // java/lang/System.in:Ljava/io/InputStream;
   #4 = Methodref          #2.#36         // java/util/Scanner."<init>":(Ljava/io/InputStream;)V
   #5 = Fieldref           #34.#37        // java/lang/System.out:Ljava/io/PrintStream;
   #6 = String             #38            // Enter num1:
   #7 = Methodref          #39.#40        // java/io/PrintStream.print:(Ljava/lang/String;)V
   #8 = Methodref          #2.#41         // java/util/Scanner.nextInt:()I
   #9 = String             #42            // Enter num2:
  #10 = String             #43            // Enter num3:
  #11 = Class              #44            // java/lang/StringBuilder
  #12 = Methodref          #11.#32        // java/lang/StringBuilder."<init>":()V
  #13 = String             #45            // The largest num is:
  #14 = Methodref          #11.#46        // java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
  #15 = Methodref          #11.#47        // java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
  #16 = Methodref          #11.#48        // java/lang/StringBuilder.toString:()Ljava/lang/String;
  #17 = Methodref          #39.#49        // java/io/PrintStream.println:(Ljava/lang/String;)V
  #18 = String             #50            // all numbers equal
  #19 = Class              #51            // largest3num
  #20 = Class              #52            // java/lang/Object
  #21 = Utf8               <init>
  #22 = Utf8               ()V
  #23 = Utf8               Code
  #24 = Utf8               LineNumberTable
  #25 = Utf8               main
  #26 = Utf8               ([Ljava/lang/String;)V
  #27 = Utf8               StackMapTable
  #28 = Class              #53            // "[Ljava/lang/String;"
  #29 = Class              #33            // java/util/Scanner
  #30 = Utf8               SourceFile
  #31 = Utf8               largest3num.java
```

```
PS D:\CDAC\assignment> javap -v largest3num
   #37 = NameAndType        #58:#59        // out:Ljava/io/PrintStream;
   #38 = Utf8               Enter num1:
   #39 = Class              #60            // java/io/PrintStream
   #40 = NameAndType        #61:#62        // print:(Ljava/lang/String;)V
   #41 = NameAndType        #63:#64        // nextInt:()I
   #42 = Utf8               Enter num2:
   #43 = Utf8               Enter num3:
   #44 = Utf8               java/lang/StringBuilder
   #45 = Utf8               The largest num is:
   #46 = NameAndType        #65:#66        // append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
   #47 = NameAndType        #65:#67        // append:(I)Ljava/lang/StringBuilder;
   #48 = NameAndType        #68:#69        // toString:()Ljava/lang/String;
   #49 = NameAndType        #70:#62        // println:(Ljava/lang/String;)V
   #50 = Utf8               all numbers equal
   #51 = Utf8               largest3num
   #52 = Utf8               java/lang/Object
   #53 = Utf8               [Ljava/lang/String;
   #54 = Utf8               java/lang/System
   #55 = Utf8               in
   #56 = Utf8               Ljava/io/InputStream;
   #57 = Utf8               (Ljava/io/InputStream;)V
   #58 = Utf8               out
   #59 = Utf8               Ljava/io/PrintStream;
   #60 = Utf8               java/io/PrintStream
   #61 = Utf8               print
   #62 = Utf8               (Ljava/lang/String;)V
   #63 = Utf8               nextInt
   #64 = Utf8               ()I
   #65 = Utf8               append
   #66 = Utf8               (Ljava/lang/String;)Ljava/lang/StringBuilder;
   #67 = Utf8               (I)Ljava/lang/StringBuilder;
   #68 = Utf8               toString
   #69 = Utf8               ()Ljava/lang/String;
   #70 = Utf8               println
{
  public largest3num();
    descriptor: ()V
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=1, args_size=1
         0: aload_0
```

```
PS D:\CDAC\assignment> javap -v largest3num
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=1, args_size=1
         0: aload_0
         1: invokespecial #1                  // Method java/lang/Object."<init>":()V
         4: return
      LineNumberTable:
        line 3: 0

  public static void main(java.lang.String[]);
    descriptor: ([Ljava/lang/String;)V
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=5, args_size=1
         0: new           #2                  // class java/util/Scanner
         3: dup
         4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
         7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
        10: astore_1
        11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        14: ldc           #6                  // String Enter num1:
        16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
        19: aload_1
        20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
        23: istore_2
        24: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        27: ldc           #9                  // String Enter num2:
        29: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
        32: aload_1
        33: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
        36: istore_3
        37: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        40: ldc           #10                 // String Enter num3:
        42: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
        45: aload_1
        46: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
        49: istore        4
        51: iload_2
        52: iload_3
        53: if_icmple     90
        56: iload_2
```

```
PS D:\CDAC\assignment> javap -v largest3num
         51: iload_2
         52: iload_3
         53: if_icmple     90
         56: iload_2
         57: iload         4
         59: if_icmple     90
         62: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
         65: new           #11                 // class java/lang/StringBuilder
         68: dup
         69: invokespecial #12                 // Method java/lang/StringBuilder."<init>":()V
         72: ldc           #13                 // String The largest num is:
         74: invokevirtual #14                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
         77: iload_2
         78: invokevirtual #15                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
         81: invokevirtual #16                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
         84: invokevirtual #17                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
         87: goto          189
         90: iload_3
         91: iload_2
         92: if_icmple     129
         95: iload_3
         96: iload         4
         98: if_icmple     129
        101: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        104: new           #11                 // class java/lang/StringBuilder
        107: dup
        108: invokespecial #12                 // Method java/lang/StringBuilder."<init>":()V
        111: ldc           #13                 // String The largest num is:
        113: invokevirtual #14                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
        116: iload_3
        117: invokevirtual #15                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
        120: invokevirtual #16                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
        123: invokevirtual #17                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
        126: goto          189
        129: iload         4
        131: iload_3
        132: if_icmple     170
        135: iload         4
        137: iload_2
        138: if_icmple     170
        141: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
```

```
PS D:\CDAC\assignment> javap -v largest3num
      138: if_icmple     170
      141: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      144: new           #11                 // class java/lang/StringBuilder
      147: dup
      148: invokespecial #12                 // Method java/lang/StringBuilder."<init>":()V
      151: ldc           #13                 // String The largest num is:
      153: invokevirtual #14                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      156: iload         4
      158: invokevirtual #15                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      161: invokevirtual #16                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      164: invokevirtual #17                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      167: goto          189
      170: iload_2
      171: iload_3
      172: if_icmpne     189
      175: iload_2
      176: iload         4
      178: if_icmpne     189
      181: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      184: ldc           #18                 // String all numbers equal
      186: invokevirtual #17                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      189: return
    LineNumberTable:
      line 6: 0
      line 7: 11
      line 8: 19
      line 9: 24
      line 10: 32
      line 10: 32
      line 11: 37
      line 12: 45
      line 14: 51
      line 12: 45
      line 14: 51
      line 14: 51
      line 15: 62
      line 16: 90
      line 17: 101
      line 18: 129
      line 19: 141
      line 21: 170
```

```
PS D:\CDAC\assignment> javap -v largest3num
        line 6: 0
        line 7: 11
        line 8: 19
        line 9: 24
        line 10: 32
        line 10: 32
        line 11: 37
        line 12: 45
        line 14: 51
        line 12: 45
        line 14: 51
        line 14: 51
        line 15: 62
        line 16: 90
        line 17: 101
        line 18: 129
        line 19: 141
        line 21: 170
        line 22: 181
        line 24: 189
      StackMapTable: number_of_entries = 4
        frame_type = 255 /* full_frame */
          offset_delta = 90
          locals = [ class "[Ljava/lang/String;", class java/util/Scanner, int, int, int ]
          stack = []
        frame_type = 38 /* same */
        frame_type = 40 /* same */
        frame_type = 18 /* same */
}
SourceFile: "largest3num.java"
PS D:\CDAC\assignment>
```

Question 7:

ASSIGNMENT

- addtwonumbers.class
- addtwonumbers.java 1
- evenorodd.class
- evenorodd.java
- factorialnum.class
- factorialnum.java
- fibonacciseries.class
- fibonacciseries.java
- largest3num.class
- largest3num.java 1
- multiplicationtable.java
- nameprint.class
- nameprint.java
- reversedigits.class
- reversedigits.java
- sumnaturalnum.java 1
- swapnum.java 1

OUTLINE
TIMELINE
JAVA PROJECTS

reversedigits.java > reversedigits

```java
import java.util.Scanner;

public class reversedigits {

    Run | Debug
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);
        System.out.print(s: "Enter num: ");

        int num = input.nextInt();
        int reversedNum = 0;

        while (num != 0) {

            int lastDigit = num % 10;
            reversedNum = (reversedNum * 10) + lastDigit;
            num = num / 10;
        }
        System.out.println("Reversed num: " + reversedNum);

        input.close();
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

```
PS D:\CDAC\assignment> javac ./reversedigits.java
PS D:\CDAC\assignment> java reversedigits
Enter num: 123497
Reversed num: 794321
PS D:\CDAC\assignment> javap reversedigits
Compiled from "reversedigits.java"
public class reversedigits {
  public reversedigits();
  public static void main(java.lang.String[]);
}
PS D:\CDAC\assignment> javap -c reversedigits
Compiled from "reversedigits.java"
public class reversedigits {
  public reversedigits();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
```

```
PS D:\CDAC\assignment> javap -c reversedigits

  public static void main(java.lang.String[]);
    Code:
       0: new           #2                  // class java/util/Scanner
       3: dup
       4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
       7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
      10: astore_1
      11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      14: ldc           #6                  // String Enter num:
      16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
      19: aload_1
      20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
      23: istore_2
      24: iconst_0
      25: istore_3
      26: iload_2
      27: ifeq          52
      30: iload_2
      31: bipush        10
      33: irem
      34: istore        4
      36: iload_3
      37: bipush        10
      39: imul
      40: iload         4
      42: iadd
      43: istore_3
      44: iload_2
      45: bipush        10
      47: idiv
      48: istore_2
      49: goto          26
      52: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      55: new           #9                  // class java/lang/StringBuilder
      58: dup
      59: invokespecial #10                 // Method java/lang/StringBuilder."<init>":()V
      62: ldc           #11                 // String Reversed num:
      64: invokevirtual #12                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      67: iload_3
      68: invokevirtual #13                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      71: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      74: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      77: aload_1
      78: invokevirtual #16                 // Method java/util/Scanner.close:()V
```

```
PS D:\CDAC\assignment> javap -v reversedigits
Classfile /D:/CDAC/assignment/reversedigits.class
  Last modified Feb 25, 2026; size 947 bytes
  MD5 checksum 661f49df8ca3ef7390179d2de1e21a5a
  Compiled from "reversedigits.java"
public class reversedigits
  minor version: 0
  major version: 52
  flags: ACC_PUBLIC, ACC_SUPER
Constant pool:
   #1 = Methodref          #18.#29        // java/lang/Object."<init>":()V
   #2 = Class              #30            // java/util/Scanner
   #3 = Fieldref           #31.#32        // java/lang/System.in:Ljava/io/InputStream;
   #4 = Methodref          #2.#33         // java/util/Scanner."<init>":(Ljava/io/InputStream;)V
   #5 = Fieldref           #31.#34        // java/lang/System.out:Ljava/io/PrintStream;
   #6 = String             #35            // Enter num:
   #7 = Methodref          #36.#37        // java/io/PrintStream.print:(Ljava/lang/String;)V
   #8 = Methodref          #2.#38         // java/util/Scanner.nextInt:()I
   #9 = Class              #39            // java/lang/StringBuilder
  #10 = Methodref          #9.#29         // java/lang/StringBuilder."<init>":()V
  #11 = String             #40            // Reversed num:
  #12 = Methodref          #9.#41         // java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
  #13 = Methodref          #9.#42         // java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
  #14 = Methodref          #9.#43         // java/lang/StringBuilder.toString:()Ljava/lang/String;
  #15 = Methodref          #36.#44        // java/io/PrintStream.println:(Ljava/lang/String;)V
  #16 = Methodref          #2.#45         // java/util/Scanner.close:()V
  #17 = Class              #46            // reversedigits
  #18 = Class              #47            // java/lang/Object
  #19 = Utf8               <init>
  #20 = Utf8               ()V
  #21 = Utf8               Code
  #22 = Utf8               LineNumberTable
  #23 = Utf8               main
  #24 = Utf8               ([Ljava/lang/String;)V
  #25 = Utf8               StackMapTable
  #26 = Class              #30            // java/util/Scanner
  #27 = Utf8               SourceFile
  #28 = Utf8               reversedigits.java
  #29 = NameAndType        #19:#20        // "<init>":()V
  #30 = Utf8               java/util/Scanner
  #31 = Class              #48            // java/lang/System
  #32 = NameAndType        #49:#50        // in:Ljava/io/InputStream;
  #33 = NameAndType        #19:#51        // "<init>":(Ljava/io/InputStream;)V
  #34 = NameAndType        #52:#53        // out:Ljava/io/PrintStream;
  #35 = Utf8               Enter num:
  #36 = Class              #54            // java/io/PrintStream
```

```
PS D:\CDAC\assignment> javap -v reversedigits
   #38 = NameAndType        #57:#58        // nextInt:()I
   #39 = Utf8               java/lang/StringBuilder
   #40 = Utf8               Reversed num:
   #41 = NameAndType        #59:#60        // append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
   #42 = NameAndType        #59:#61        // append:(I)Ljava/lang/StringBuilder;
   #43 = NameAndType        #62:#63        // toString:()Ljava/lang/String;
   #44 = NameAndType        #64:#56        // println:(Ljava/lang/String;)V
   #45 = NameAndType        #65:#20        // close:()V
   #46 = Utf8               reversedigits
   #47 = Utf8               java/lang/Object
   #48 = Utf8               java/lang/System
   #49 = Utf8               in
   #50 = Utf8               Ljava/io/InputStream;
   #51 = Utf8               (Ljava/io/InputStream;)V
   #52 = Utf8               out
   #53 = Utf8               Ljava/io/PrintStream;
   #54 = Utf8               java/io/PrintStream
   #55 = Utf8               print
   #56 = Utf8               (Ljava/lang/String;)V
   #57 = Utf8               nextInt
   #58 = Utf8               ()I
   #59 = Utf8               append
   #60 = Utf8               (Ljava/lang/String;)Ljava/lang/StringBuilder;
   #61 = Utf8               (I)Ljava/lang/StringBuilder;
   #62 = Utf8               toString
   #63 = Utf8               ()Ljava/lang/String;
   #64 = Utf8               println
   #65 = Utf8               close
{
  public reversedigits();
    descriptor: ()V
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=1, args_size=1
         0: aload_0
         1: invokespecial #1                  // Method java/lang/Object."<init>":()V
         4: return
      LineNumberTable:
        line 3: 0

  public static void main(java.lang.String[]);
    descriptor: ([Ljava/lang/String;)V
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=5, args_size=1
```

```
PS D:\CDAC\assignment> javap -v reversedigits
         0: new           #2                  // class java/util/Scanner
         3: dup
         4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
         7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
        10: astore_1
        11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        14: ldc           #6                  // String Enter num:
        16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
        19: aload_1
        20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
        23: istore_2
        24: iconst_0
        25: istore_3
        26: iload_2
        27: ifeq          52
        30: iload_2
        31: bipush        10
        33: irem
        34: istore        4
        36: iload_3
        37: bipush        10
        39: imul
        40: iload         4
        42: iadd
        43: istore_3
        44: iload_2
        45: bipush        10
        47: idiv
        48: istore_2
        49: goto          26
        52: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        55: new           #9                  // class java/lang/StringBuilder
        58: dup
        59: invokespecial #10                 // Method java/lang/StringBuilder."<init>":()V
        62: ldc           #11                 // String Reversed num:
        64: invokevirtual #12                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
        67: iload_3
        68: invokevirtual #13                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
        71: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
        74: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
        77: aload_1
        78: invokevirtual #16                 // Method java/util/Scanner.close:()V
        81: return
      LineNumberTable:
        74: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
```

```
PS D:\CDAC\assignment> javap -v reversedigits
        77: aload_1
        78: invokevirtual #16                 // Method java/util/Scanner.close:()V
        81: return
      LineNumberTable:
        74: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
        77: aload_1
        78: invokevirtual #16                 // Method java/util/Scanner.close:()V
        81: return
      LineNumberTable:
        77: aload_1
        78: invokevirtual #16                 // Method java/util/Scanner.close:()V
        81: return
      LineNumberTable:
        78: invokevirtual #16                 // Method java/util/Scanner.close:()V
        81: return
      LineNumberTable:
        81: return
      LineNumberTable:
      LineNumberTable:
        line 7: 0
        line 8: 11
        line 10: 19
        line 11: 24
        line 11: 24
        line 13: 26
        line 15: 30
        line 16: 36
        line 17: 44
        line 18: 49
        line 19: 52
        line 21: 77
        line 22: 81
      StackMapTable: number_of_entries = 2
        frame_type = 254 /* append */
          offset_delta = 26
          locals = [ class java/util/Scanner, int, int ]
        frame_type = 25 /* same */
}
SourceFile: "reversedigits.java"
PS D:\CDAC\assignment>
```

Question 8:

ASSIGNMENT

- addtwonumbers.class
- addtwonumbers.java 1
- evenorodd.class
- evenorodd.java
- factorialnum.class
- factorialnum.java
- fibonacciseries.class
- fibonacciseries.java
- largest3num.class
- largest3num.java 1
- multiplicationtable.class
- multiplicationtable.java
- nameprint.class
- nameprint.java
- reversedigits.class
- reversedigits.java
- sumnaturalnum.java 1
- swapnum.java 1

OUTLINE
TIMELINE
JAVA PROJECTS

multiplicationtable.java > multiplicationtable > main(String[])

```java
import java.util.Scanner;

public class multiplicationtable {
    Run | Debug
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print(s: "Enter num: ");
        int num = input.nextInt();

        for (int i = 1; i <= 10; i++) {
            int result = num * i;
            System.out.println(num + " x " + i + " = " + result);
        }

        input.close();
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell

```
PS D:\CDAC\assignment>
```

```
PS D:\CDAC\assignment> javac ./multiplicationtable.java
PS D:\CDAC\assignment> java multiplicationtable
Enter num: 9
9 x 1 = 9
9 x 2 = 18
9 x 3 = 27
9 x 4 = 36
9 x 5 = 45
9 x 6 = 54
9 x 7 = 63
9 x 8 = 72
9 x 9 = 81
9 x 10 = 90
PS D:\CDAC\assignment> javap multiplicationtable
Compiled from "multiplicationtable.java"
public class multiplicationtable {
  public multiplicationtable();
```

```
PS D:\CDAC\assignment> javap -c multiplicationtable
      10: astore_1
      11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      14: ldc           #6                  // String Enter num:
      16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
      19: aload_1
      20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
      23: istore_2
      24: iconst_1
      25: istore_3
      26: iload_3
      27: bipush        10
      29: if_icmpgt     82
      32: iload_2
      33: iload_3
      34: imul
      35: istore        4
      37: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      40: new           #9                  // class java/lang/StringBuilder
      43: dup
      44: invokespecial #10                 // Method java/lang/StringBuilder."<init>":()V
      47: iload_2
      48: invokevirtual #11                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      51: ldc           #12                 // String  x
      53: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      56: iload_3
      57: invokevirtual #11                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      60: ldc           #14                 // String  =
      62: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      65: iload         4
      67: invokevirtual #11                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      70: invokevirtual #15                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      73: invokevirtual #16                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      76: iinc          3, 1
      79: goto          26
      82: aload_1
      83: invokevirtual #17                 // Method java/util/Scanner.close:()V
      86: return
}
```

```
PS D:\CDAC\assignment> javap -c multiplicationtable
}
PS D:\CDAC\assignment> javap -v multiplicationtable
Classfile /D:/CDAC/assignment/multiplicationtable.class
  Last modified Feb 25, 2026; size 952 bytes
  MD5 checksum bb2bff0fd63691757cc937ae41b631e5
  Compiled from "multiplicationtable.java"
public class multiplicationtable
  minor version: 0
  major version: 52
  flags: ACC_PUBLIC, ACC_SUPER
Constant pool:
   #1 = Methodref          #19.#30        // java/lang/Object."<init>":()V
   #2 = Class              #31            // java/util/Scanner
   #3 = Fieldref           #32.#33        // java/lang/System.in:Ljava/io/InputStream;
   #4 = Methodref          #2.#34         // java/util/Scanner."<init>":(Ljava/io/InputStream;)V
   #5 = Fieldref           #32.#35        // java/lang/System.out:Ljava/io/PrintStream;
   #6 = String             #36            // Enter num:
   #7 = Methodref          #37.#38        // java/io/PrintStream.print:(Ljava/lang/String;)V
   #8 = Methodref          #2.#39         // java/util/Scanner.nextInt:()I
   #9 = Class              #40            // java/lang/StringBuilder
  #10 = Methodref          #9.#30         // java/lang/StringBuilder."<init>":()V
  #11 = Methodref          #9.#41         // java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
  #12 = String             #42            //  x
  #13 = Methodref          #9.#43         // java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
  #14 = String             #44            //  =
  #15 = Methodref          #9.#45         // java/lang/StringBuilder.toString:()Ljava/lang/String;
  #16 = Methodref          #37.#46        // java/io/PrintStream.println:(Ljava/lang/String;)V
  #17 = Methodref          #2.#47         // java/util/Scanner.close:()V
  #18 = Class              #48            // multiplicationtable
  #19 = Class              #49            // java/lang/Object
  #20 = Utf8               <init>
  #21 = Utf8               ()V
  #22 = Utf8               Code
  #23 = Utf8               LineNumberTable
  #24 = Utf8               main
  #25 = Utf8               ([Ljava/lang/String;)V
  #26 = Utf8               StackMapTable
  #27 = Class              #31            // java/util/Scanner
  #28 = Utf8               SourceFile
  #29 = Utf8               multiplicationtable.java
```

```
PS D:\CDAC\assignment> javap -v multiplicationtable
  #58 = Utf8               (Ljava/lang/String;)V
  #59 = Utf8               nextInt
  #60 = Utf8               ()I
  #61 = Utf8               append
  #62 = Utf8               (I)Ljava/lang/StringBuilder;
  #63 = Utf8               (Ljava/lang/String;)Ljava/lang/StringBuilder;
  #64 = Utf8               toString
  #65 = Utf8               ()Ljava/lang/String;
  #66 = Utf8               println
  #67 = Utf8               close
{
  public multiplicationtable();
    descriptor: ()V
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=1, args_size=1
         0: aload_0
         1: invokespecial #1                  // Method java/lang/Object."<init>":()V
         4: return
      LineNumberTable:
        line 3: 0

  public static void main(java.lang.String[]);
    descriptor: ([Ljava/lang/String;)V
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=5, args_size=1
         0: new           #2                  // class java/util/Scanner
         3: dup
         4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
         7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
        10: astore_1
        11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        14: ldc           #6                  // String Enter num:
        16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
        19: aload_1
        20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
        23: istore_2
        24: iconst_1
        25: istore_3
```

```
PS D:\CDAC\assignment> javap -v multiplicationtable
        32: iload_2
        33: iload_3
        34: imul
        35: istore        4
        37: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        40: new           #9                  // class java/lang/StringBuilder
        43: dup
        44: invokespecial #10                 // Method java/lang/StringBuilder."<init>":()V
        47: iload_2
        48: invokevirtual #11                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
        51: ldc           #12                 // String  x
        53: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
        56: iload_3
        57: invokevirtual #11                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
        60: ldc           #14                 // String  =
        62: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
        65: iload         4
        67: invokevirtual #11                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
        70: invokevirtual #15                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
        73: invokevirtual #16                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
        76: iinc          3, 1
        79: goto          26
        82: aload_1
        83: invokevirtual #17                 // Method java/util/Scanner.close:()V
        86: return
      LineNumberTable:
        line 5: 0
        line 7: 11
        line 8: 19
        line 10: 24
        line 11: 32
        line 12: 37
        line 10: 76
        line 15: 82
        line 16: 86
      StackMapTable: number_of_entries = 2
        frame_type = 254 /* append */
          offset_delta = 26
          locals = [ class java/util/Scanner, int, int ]
        frame_type = 250 /* chop */
          offset_delta = 55
}
SourceFile: "multiplicationtable.java"
```

Question 9:

EXPLORER

ASSIGNMENT

- addtwonumbers.class
- addtwonumbers.java 1
- evenorodd.class
- evenorodd.java
- factorialnum.class
- factorialnum.java
- fibonacciseries.class
- fibonacciseries.java
- largest3num.class
- largest3num.java 1
- multiplicationtable.class
- multiplicationtable.java
- nameprint.class
- nameprint.java
- reversedigits.class
- reversedigits.java
- sumnaturalnum.class
- sumnaturalnum.java 1
- swapnum.java 1

sumnaturalnum.java 1

sumnaturalnum.java > sumnaturalnum

```java
import java.util.Scanner;

public class sumnaturalnum {
    Run | Debug
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print(s: "Enter a num: ");
        int n = input.nextInt();

        int sum = 0;
        for (int i = 1; i <= n; i++) {
            sum = sum + i;
        }
        System.out.println("Sum: " + sum);
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CDAC\assignment> javac ./sumnaturalnum.java
PS D:\CDAC\assignment> java sumnaturalnum
Enter a num: 12
Sum: 78
PS D:\CDAC\assignment>
```

```
PS D:\CDAC\assignment> javap -c sumnaturalnum
Compiled from "sumnaturalnum.java"
public class sumnaturalnum {
  public sumnaturalnum();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: new           #2                  // class java/util/Scanner
       3: dup
       4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
       7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
      10: astore_1
      11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      14: ldc           #6                  // String Enter a num:
      16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
      19: aload_1
      20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
      23: istore_2
      24: iconst_0
      25: istore_3
      26: iconst_1
      27: istore        4
      29: iload         4
      31: iload_2
      32: if_icmpgt     46
      35: iload_3
      36: iload         4
      38: iadd
      39: istore_3
      40: iinc          4, 1
      43: goto          29
      46: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      49: new           #9                  // class java/lang/StringBuilder
```

```
PS D:\CDAC\assignment> javap -v sumnaturalnum
Classfile /D:/CDAC/assignment/sumnaturalnum.class
  Last modified Feb 25, 2026; size 935 bytes
  MD5 checksum 9d50c5e0f86b47e57c0d02b5201b85ae
  Compiled from "sumnaturalnum.java"
public class sumnaturalnum
  minor version: 0
  major version: 52
  flags: ACC_PUBLIC, ACC_SUPER
Constant pool:
   #1 = Methodref          #17.#29        // java/lang/Object."<init>":()V
   #2 = Class              #30            // java/util/Scanner
   #3 = Fieldref           #31.#32        // java/lang/System.in:Ljava/io/InputStream;
   #4 = Methodref          #2.#33         // java/util/Scanner."<init>":(Ljava/io/InputStream;)V
   #5 = Fieldref           #31.#34        // java/lang/System.out:Ljava/io/PrintStream;
   #6 = String             #35            // Enter a num:
   #7 = Methodref          #36.#37        // java/io/PrintStream.print:(Ljava/lang/String;)V
   #8 = Methodref          #2.#38         // java/util/Scanner.nextInt:()I
   #9 = Class              #39            // java/lang/StringBuilder
  #10 = Methodref          #9.#29         // java/lang/StringBuilder."<init>":()V
  #11 = String             #40            // Sum:
  #12 = Methodref          #9.#41         // java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
  #13 = Methodref          #9.#42         // java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
  #14 = Methodref          #9.#43         // java/lang/StringBuilder.toString:()Ljava/lang/String;
  #15 = Methodref          #36.#44        // java/io/PrintStream.println:(Ljava/lang/String;)V
  #16 = Class              #45            // sumnaturalnum
  #17 = Class              #46            // java/lang/Object
  #18 = Utf8               <init>
  #19 = Utf8               ()V
  #20 = Utf8               Code
  #21 = Utf8               LineNumberTable
  #22 = Utf8               main
  #23 = Utf8               ([Ljava/lang/String;)V
  #24 = Utf8               StackMapTable
  #25 = Class              #47            // "[Ljava/lang/String;"
  #26 = Class              #30            // java/util/Scanner
  #27 = Utf8               SourceFile
  #28 = Utf8               sumnaturalnum.java
```

```
PS D:\CDAC\assignment> javap -v sumnaturalnum
  #63 = Utf8               ()Ljava/lang/String;
  #64 = Utf8               println
{
  public sumnaturalnum();
    descriptor: ()V
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=1, args_size=1
         0: aload_0
         1: invokespecial #1                  // Method java/lang/Object."<init>":()V
         4: return
      LineNumberTable:
        line 3: 0

  public static void main(java.lang.String[]);
    descriptor: ([Ljava/lang/String;)V
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=5, args_size=1
         0: new           #2                  // class java/util/Scanner
         3: dup
         4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
         7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
        10: astore_1
        11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        14: ldc           #6                  // String Enter a num:
        16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
        19: aload_1
        20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
        23: istore_2
        24: iconst_0
        25: istore_3
        26: iconst_1
        27: istore        4
        29: iload         4
        31: iload_2
        32: if_icmpgt     46
        35: iload_3
        36: iload         4
        38: iadd
```

```
PS D:\CDAC\assignment> javap -v sumnaturalnum
        39: istore_3
        40: iinc          4, 1
        43: goto          29
        46: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        49: new           #9                  // class java/lang/StringBuilder
        52: dup
        53: invokespecial #10                 // Method java/lang/StringBuilder."<init>":()V
        56: ldc           #11                 // String Sum:
        58: invokevirtual #12                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
        61: iload_3
        62: invokevirtual #13                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
        65: invokevirtual #14                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
        68: invokevirtual #15                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
        71: return
      LineNumberTable:
        line 5: 0
        line 7: 11
        line 8: 19
        line 10: 24
        line 11: 26
        line 12: 35
        line 11: 40
        line 14: 46
        line 15: 71
      StackMapTable: number_of_entries = 2
        frame_type = 255 /* full_frame */
          offset_delta = 29
          locals = [ class "[Ljava/lang/String;", class java/util/Scanner, int, int, int ]
          stack = []
        frame_type = 250 /* chop */
          offset_delta = 16
}
SourceFile: "sumnaturalnum.java"
PS D:\CDAC\assignment>
```

Question 10:

ASSIGNMENT

- addtwonumbers.class
- addtwonumbers.java 1
- evenorodd.class
- evenorodd.java
- factorialnum.class
- factorialnum.java
- fibonacciseries.class
- fibonacciseries.java
- largest3num.class
- largest3num.java 1
- multiplicationtable.class
- multiplicationtable.java
- nameprint.class
- nameprint.java
- reversedigits.class
- reversedigits.java
- sumnaturalnum.class
- sumnaturalnum.java 1
- swapnum.class
- swapnum.java 1

swapnum.java > swapnum

```java
import java.util.Scanner;

public class swapnum {
    Run | Debug
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.print(s: "Enter num1: ");
        int n1 = input.nextInt();

        System.out.print(s: "Enter num2: ");
        int n2 = input.nextInt();

        n1 = n1 + n2;
        n2 = n1 - n2;
        n1 = n1 - n2;

        System.out.println("num1 = " + n1 + ", num2 = " + n2);
    }
}
```

PROBLEMS 4 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\CDAC\assignment> javac ./swapnum.java
PS D:\CDAC\assignment> java swapnum
Enter num1: 13
Enter num2: 24
num1 = 24, num2 = 13
PS D:\CDAC\assignment>
```

```
PS D:\CDAC\assignment> javap -c swapnum
Compiled from "swapnum.java"
public class swapnum {
  public swapnum();
    Code:
       0: aload_0
       1: invokespecial #1                  // Method java/lang/Object."<init>":()V
       4: return

  public static void main(java.lang.String[]);
    Code:
       0: new           #2                  // class java/util/Scanner
       3: dup
       4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
       7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
      10: astore_1
      11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      14: ldc           #6                  // String Enter num1:
      16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
      19: aload_1
      20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
      23: istore_2
      24: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      27: ldc           #9                  // String Enter num2:
      29: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
      32: aload_1
      33: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
      36: istore_3
      37: iload_2
      38: iload_3
      39: iadd
      40: istore_2
      41: iload_2
      42: iload_3
      43: isub
      44: istore_3
      45: iload_2
      46: iload_3
      47: isub
      48: istore_2
```

```
PS D:\CDAC\assignment> javap -c swapnum
      41: iload_2
      42: iload_3
      43: isub
      44: istore_3
      45: iload_2
      46: iload_3
      47: isub
      48: istore_2
      49: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
      52: new           #10                 // class java/lang/StringBuilder
      55: dup
      56: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
      59: ldc           #12                 // String num1 =
      61: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      64: iload_2
      65: invokevirtual #14                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      68: ldc           #15                 // String , num2 =
      70: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
      73: iload_3
      74: invokevirtual #14                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
      77: invokevirtual #16                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
      80: invokevirtual #17                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
      83: return
}
PS D:\CDAC\assignment> javap -v swapnum
Classfile /D:/CDAC/assignment/swapnum.class
  Last modified Feb 25, 2026; size 902 bytes
  MD5 checksum 03ed8b2ab854691a28347c3feaefa783
  Compiled from "swapnum.java"
public class swapnum
  minor version: 0
  major version: 52
  flags: ACC_PUBLIC, ACC_SUPER
Constant pool:
   #1 = Methodref          #19.#28        // java/lang/Object."<init>":()V
   #2 = Class              #29            // java/util/Scanner
   #3 = Fieldref           #30.#31        // java/lang/System.in:Ljava/io/InputStream;
   #4 = Methodref          #2.#32         // java/util/Scanner."<init>":(Ljava/io/InputStream;)V
   #5 = Fieldref           #30.#33        // java/lang/System.out:Ljava/io/PrintStream;
```

```
PS D:\CDAC\assignment> javap -v swapnum
   #13 = Methodref          #10.#41        // java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
   #14 = Methodref          #10.#42        // java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
   #15 = String             #43            // , num2 =
   #16 = Methodref          #10.#44        // java/lang/StringBuilder.toString:()Ljava/lang/String;
   #17 = Methodref          #35.#45        // java/io/PrintStream.println:(Ljava/lang/String;)V
   #18 = Class              #46            // swapnum
   #19 = Class              #47            // java/lang/Object
   #20 = Utf8               <init>
   #21 = Utf8               ()V
   #22 = Utf8               Code
   #23 = Utf8               LineNumberTable
   #24 = Utf8               main
   #25 = Utf8               ([Ljava/lang/String;)V
   #26 = Utf8               SourceFile
   #27 = Utf8               swapnum.java
   #28 = NameAndType        #20:#21        // "<init>":()V
   #29 = Utf8               java/util/Scanner
   #30 = Class              #48            // java/lang/System
   #31 = NameAndType        #49:#50        // in:Ljava/io/InputStream;
   #32 = NameAndType        #20:#51        // "<init>":(Ljava/io/InputStream;)V
   #33 = NameAndType        #52:#53        // out:Ljava/io/PrintStream;
   #34 = Utf8               Enter num1:
   #35 = Class              #54            // java/io/PrintStream
   #36 = NameAndType        #55:#56        // print:(Ljava/lang/String;)V
   #37 = NameAndType        #57:#58        // nextInt:()I
   #38 = Utf8               Enter num2:
   #39 = Utf8               java/lang/StringBuilder
   #40 = Utf8               num1 =
   #41 = NameAndType        #59:#60        // append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
   #42 = NameAndType        #59:#61        // append:(I)Ljava/lang/StringBuilder;
   #43 = Utf8               , num2 =
   #44 = NameAndType        #62:#63        // toString:()Ljava/lang/String;
   #45 = NameAndType        #64:#56        // println:(Ljava/lang/String;)V
   #46 = Utf8               swapnum
   #47 = Utf8               java/lang/Object
   #48 = Utf8               java/lang/System
   #49 = Utf8               in
   #50 = Utf8               Ljava/io/InputStream;
   #51 = Utf8               (Ljava/io/InputStream;)V
```

```
PS D:\CDAC\assignment> javap -v swapnum
  #64 = Utf8               println
{
  public swapnum();
    descriptor: ()V
    flags: ACC_PUBLIC
    Code:
      stack=1, locals=1, args_size=1
         0: aload_0
         1: invokespecial #1                  // Method java/lang/Object."<init>":()V
         4: return
      LineNumberTable:
        line 3: 0

  public static void main(java.lang.String[]);
    descriptor: ([Ljava/lang/String;)V
    flags: ACC_PUBLIC, ACC_STATIC
    Code:
      stack=3, locals=4, args_size=1
         0: new           #2                  // class java/util/Scanner
         3: dup
         4: getstatic     #3                  // Field java/lang/System.in:Ljava/io/InputStream;
         7: invokespecial #4                  // Method java/util/Scanner."<init>":(Ljava/io/InputStream;)V
        10: astore_1
        11: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        14: ldc           #6                  // String Enter num1:
        16: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
        19: aload_1
        20: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
        23: istore_2
        24: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
        27: ldc           #9                  // String Enter num2:
        29: invokevirtual #7                  // Method java/io/PrintStream.print:(Ljava/lang/String;)V
        32: aload_1
        33: invokevirtual #8                  // Method java/util/Scanner.nextInt:()I
        36: istore_3
        37: iload_2
        38: iload_3
        39: iadd
        40: istore_2
```

```
PS D:\CDAC\assignment> javap -v swapnum
         39: iadd
         40: istore_2
         41: iload_2
         42: iload_3
         43: isub
         44: istore_3
         45: iload_2
         46: iload_3
         47: isub
         48: istore_2
         49: getstatic     #5                  // Field java/lang/System.out:Ljava/io/PrintStream;
         52: new           #10                 // class java/lang/StringBuilder
         55: dup
         56: invokespecial #11                 // Method java/lang/StringBuilder."<init>":()V
         59: ldc           #12                 // String num1 =
         61: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
         64: iload_2
         65: invokevirtual #14                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
         68: ldc           #15                 // String , num2 =
         70: invokevirtual #13                 // Method java/lang/StringBuilder.append:(Ljava/lang/String;)Ljava/lang/StringBuilder;
         73: iload_3
         74: invokevirtual #14                 // Method java/lang/StringBuilder.append:(I)Ljava/lang/StringBuilder;
         77: invokevirtual #16                 // Method java/lang/StringBuilder.toString:()Ljava/lang/String;
         80: invokevirtual #17                 // Method java/io/PrintStream.println:(Ljava/lang/String;)V
         83: return
       LineNumberTable:
         line 5: 0
         line 7: 11
         line 8: 19
         line 10: 24
         line 11: 32
         line 13: 37
         line 14: 41
         line 15: 45
         line 17: 49
         line 18: 83
}
SourceFile: "swapnum.java"
PS D:\CDAC\assignment>
```